



CUSTOMER RETENTION STRATEGY

CAPSTONE PROJECT OVERVIEW – “AURA HOME SYSTEMS CUSTOMER RETENTION”

PROJECT REQUIREMENTS

Project Objective: Translate major elements learned pertaining to the degree program into a real-world actionable assessment. Implement an app that applies to your project that also has real-world implications.

Required Tools:

- ✓ Excel
- ✓ SQL
- ✓ Google Colab/Python
- ✓ Tableau or Power BI
- ✓ Github
- ✓ Presentation Software
- ✓ Vibe-Coding Application Creation

Note: Specifics on HOW to use these tools are subjective to the context of the project. Also, at the end of the presentation, there are visuals of the dashboards and the app for reference.

PROJECT DETAILS

Dataset: Aura Home Systems customer dataset

Goal: Customer retention strategies based on customer segmentation

Techniques:

- ✓ Data cleaning in Excel
- ✓ Machine Learning in Google Colab/Python (Random Forests and K Means Clustering)
- ✓ Querying data and additional segmentation in SQL
- ✓ Visualizing data in Power BI
- ✓ Application creation using Windsurf
- ✓ Dedicated Github repository for this project



CUSTOMER RETENTION STRATEGY

Presented By: David Perez
Presented On: July 31, 2025

Agenda Diagram





**RETENTION
BACKGROUND**

ABOUT AURA



Aura Home Systems is a connectivity platform with services including phone, internet, and online security services, as well as online television and movie streaming.

AURA BY THE NUMBERS (PART 1)



- Aura has lost 26.54% of its customer base valued at \$139K
- Aura currently generates \$456K in monthly recurring charges
- Aura currently has an at-risk revenue loss potential of \$100K in monthly recurring charges

AURA BY THE NUMBERS (PART 2)



- Aura has an average churn probability of 27.53%
- The average Aura customer generates \$64.76/month
- The average customer lifetime value of an Aura customer is \$169.60

RETENTION BY THE NUMBERS (PART 1)



- It is 5-25x more expensive to acquire new customers than to retain one
- A 5% retention rate can generate 25-95% more profit
- An existing customer has a 60-70% chance of buying again compared to 5-20% for a new prospect

RETENTION BY THE NUMBERS (PART 2)



- Companies with a retention rate higher than 85% grow 1.8x faster
- ROI on retention is on average 5x higher than on acquisition marketing measured over a 24-month period

AURA VS INDUSTRY STANDARDS



- The average churn rate for a telecom company is 22.8%. Aura has a current churn rate of 26.54%.
- The average gross revenue retention rate for a telecom company is between 80-88%. Aura currently has a GRR of 69.50%

RETENTION SUMMARY



Compared to industry standards, Aura is under-performing when it comes to churning customers and retaining gross revenue. Given that customer retention seems to equate to a healthier business model, we feel we can implement a customer retention strategy that can be beneficial for the company.



GOAL & METHODOLOGY

THE GOAL



Our goal is to retain as many customers as possible. It is the most cost-effective ideology in retaining recurring revenue and has the potential to generate additional revenue as well.

CHURN PROBABILITY



According to industry standards, customers with a $<30\%$ churn probability are very unlikely to churn and customers with a $>90\%$ churn probability are very likely to churn and are considered “lost causes.”

THE METHODOLOGY



- Allocate resources to current customers who have a 30-90% chance of churning
- Assessing the churn probability and at-risk value of each customer
- Segmenting customers based on their value and churn probability using machine learning algorithms
- Targeting segmented tiers with customized incentives for each respective tier

CUSTOMER SEGMENTATION



- **Tier 1:**
Low Value / High Risk +
Low Value / Med Risk
- **Tier 2:**
High Value / Med Risk +
Med Value / Med Risk
- **Tier 3:**
High Value / High Risk +
Med Value / High Risk

“TIER 1” BY THE NUMBERS



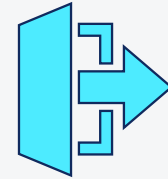
Number of
Customers
255



At-Risk
Revenue
\$3,100



Max Potential
MRR Loss
\$6,523



Avg Churn
Probability
48%



Min Monthly
Charges
\$18.85



Max Monthly
Charges
\$42.40



Average Revenue
Per User
\$25.58



Avg Customer
Lifetime Value
\$59.96

“TIER 2” BY THE NUMBERS



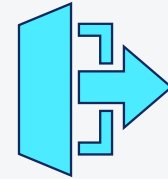
Number of
Customers
1,467



At-Risk
Revenue
\$58,963



Max Potential
MRR Loss
\$118,166



Avg Churn
Probability
50%



Min Monthly
Charges
\$43.35



Max Monthly
Charges
\$118.60



Average Revenue
Per User
\$80.55



Avg Customer
Lifetime Value
\$231.36

“TIER 3” BY THE NUMBERS



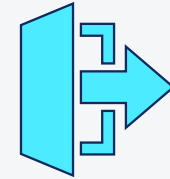
Number of
Customers
232



At-Risk
Revenue
\$15,194



Max Potential
MRR Loss
\$18,772



Avg Churn
Probability
81%



Min Monthly
Charges
\$44.30



Max Monthly
Charges
\$111.40



Average Revenue
Per User
\$80.91



Avg Customer
Lifetime Value
\$110.38

RETENTION IDEOLOGY

“
Incentivize
&
Minimize
”

- ✓ Incentivize customers with minimal resources
- ✓ Incentivizing customers will minimize churning

RETENTION CAMPAIGN COST

- The three targeted tiers generate a total monthly recurring revenue amount of \$143K
- Monthly Recurring Revenue x 7% = Marketing Budget
- Marketing Budget x 40% = Campaign Cost
- $(\text{MRR} \times 7\%) \times 40\% = \text{Campaign Cost (Alternative Formula)}$
- **$(\$143,461 \times 7\%) \times 40\% = \sim\$4,017 \text{ for Campaign Costs}$**



METHODOLOGY SUMMARY



Segmenting customers into tiers based on their churn probability and monetary value will allow us to strategically allocate resources to retaining “at-risk” yet “retainable” customers. We will be using our “incentivize and minimize” ideology to map-out a customized strategy for each respective tier.



UNIVERSAL RETENTION STRATEGY

UNIVERSAL RETENTION PREFACE



It is important that we acknowledge universal retention methods that can assist us in our long-term goals of retaining customers as they tie into our tier-based retention strategy.

UNIVERSAL STRATEGY IDEOLOGY



Minimize spending where applicable while motivating customers to stay with Aura using incentives that are at little or no real cost to the company.

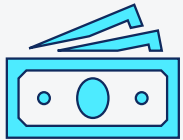
UNIVERSAL INCENTIVES



Referral program with a \$25 bill credit for each successful referral



Customer out-reach to evaluate the quality of their experience on a periodic basis



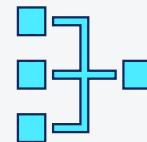
\$5 monthly credit for automated payments



Free equipment cleaning and diagnostic check



Personalized “Thank You” message



Loyalty, points-based rewards program

LOYALTY PROGRAMS BY THE NUMBERS

- 67% of members spend more vs non-members
- 28% higher purchasing frequency among members
- 79% of customers say loyalty programs influence their choices
- 77% of customers say a good loyalty program makes them loyal to a brand even if a competitor is cheaper
- **Overall loyalty program success rate of 84%**



UNIVERSAL STRATEGY SUMMARY



Creating universal incentives reinforces retention capabilities and assists us with our granular retention strategy. Implementing a points-based loyalty program can increase customer interaction, thus strengthening the relationship between Aura and its customers, as well as fortify positive sentiments towards the company.



TIER-BASED STRATEGY

TIER IDEOLOGY

Tier

1

**Low Value/High Risk +
Low Value/Med Risk**

Low value so incentives
must be low cost and
scalable.

Tier

2

**High Value/Med Risk +
Med Value/Med Risk**

Not in immediate
danger but are still high
value.

Tier

3

**High Value/High Risk +
Med Value/High Risk**

High churn probability
and meaningful impact.

“TIER 1” STRATEGY



Low cost and scalable incentives for our Tier 1 customers:

- \$5 off one-time bill credit
- Free month of a free streaming service add-on
- Prepaid 6-month bundle options
- Free Modem upgrade
- Free Wi-fi extender

“TIER 2” STRATEGY



All Tier 1 incentives plus automatic enrollment of the customer loyalty program that includes points for the following:

- On-time payments
- Referrals
- Autopay enrollment
- Downloading the app
- Completing a customer profile
- Participating in surveys

Points can be redeemed for bill credits and/or upgrades.

“TIER 3” STRATEGY



Customers in this tier group need personalized outreach! This tier includes all incentives from Tiers 1 and 2 plus the following:

- Priority customer support
- Faster response service
- Early access to new plans
- Free month of service after 12 months
- Waived activation/upgrade fees
- Free add-on streaming service for 3-6 months
- Data cap increase

ADDITIONAL STRATEGY

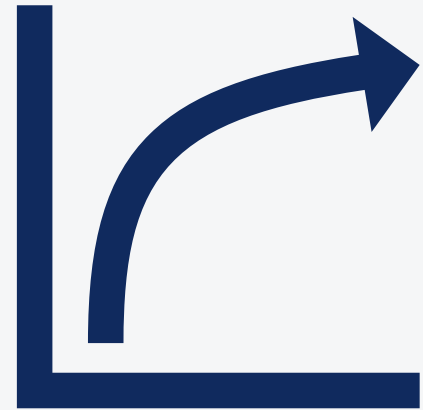


Within these tiers, newer month-to-month customers are the highest “at-risk.” We could offer the following to retain and transition them into a more solidified contract structure:

- Plan optimization suggestions
- Cheaper plan alternatives with an extended commitment
- Add-on bundles
- 5% off annual plan

TIER RETENTION BY THE NUMBERS

- Total “tier” max Recoverable Revenue is valued at \$143K
- (Max Recoverable Revenue x 25%) – Campaign Costs = Expected Recoverable Revenue
- ($\sim \$143\text{K} \times \sim 25\%$) - $\sim \$4\text{K}$ = **$\sim \$32\text{K}$ Expected Recoverable Revenue**
- Expected Recoverable Revenue / Campaign Cost = Retention ROI
- **$\sim \$32\text{K} / \sim \$4\text{K} = \sim 793\%$ Retention ROI**



TIER-BASED STRATEGY SUMMARY



By segmenting “at-risk” and “retainable” customers into tiers, we can strategically allocate resources based on the customers monetary value and at-risk probability combination. In doing so we predict a minimum \$32K retention value with a \$4K retention investment, giving us a retention ROI of 793%.



CONCLUSION & APPENDIX

CONCLUSION

The importance of customer retention is evident. It is cheaper to retain customers than it is to acquire new ones, loyal customers are more likely to spend more with the company, and retaining customers provides more consistent revenue-based stability.

By segmenting “retainable” and “at-risk” customers we can focus our retention efforts appropriately.



NEXT STEPS



Interdepartmental meeting with accounting, marketing, and staff managers to further discuss the cost particulars of the retention strategy and how to allocate the campaign-cost budget effectively.

FURTHER RESEARCH

- Date specifics for the dataset. Having an idea of the time of year when customers are most likely to churn can help us understand seasonal churning to further strengthen our retention strategy.
- Name, address, and email information for each customer included in the dataset. This can streamline multi-department usage.
- Customer address information can also help us see if there are any regional patterns that we need to watch out for.
- Preferred method of contact for each customer to further understand customer behavior when assessing at-risk potential.
- Obtaining this data would allow me to create an app that can be a “one-stop shop” for multi-departmental usage.

REFERENCES

- **Retention statistics source**

<https://searchlab.nl/en/statistics/customer-retention-statistics-2026>

- **Marketing budget sources**

<https://www.bigcommerce.com/glossary/marketing-budget/>

<https://evokad.com/marketing-budget-guide-customer-retention-acquisition/>

- **Retention metrics and formulas sources**

<https://customergauge.com/customer-retention>

<https://www.dock.us/library/customer-success-metrics>

- **Retention strategy sources**

<https://paylode.com/articles/how-telecom-brands-can-slash-churn-without-slashing-prices>

<https://propellocloud.com/blog/telecom-loyalty-programs/>

<https://propellocloud.com/blog/tiered-loyalty-program/>

“NUMBERS AT A GLANCE” DASHBOARD

A glimpse of the company with metrics applicable to customer retention

Aura Home Systems - Numbers at a Glance

Churn Rate %

26.54%

Total MRR

\$456,117

Avg CLV

\$169.60

Churn Probability

27.53%

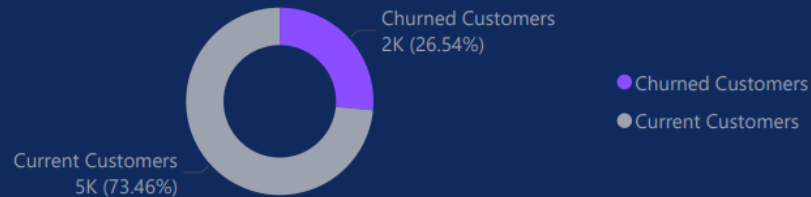
Total Customers

7,043

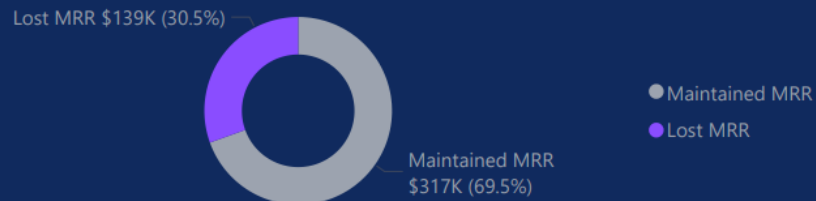
ARPU

\$64.76

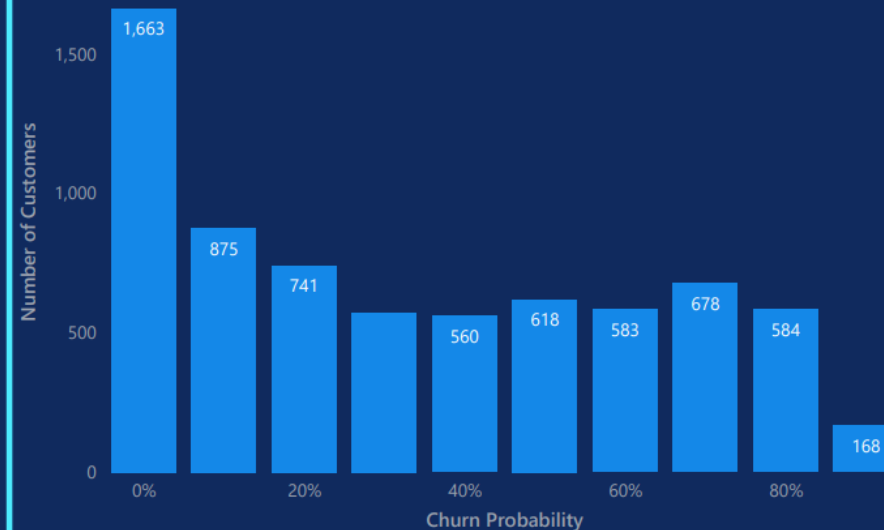
Churned Customers vs Current Customers



Lost MRR vs Maintained MRR

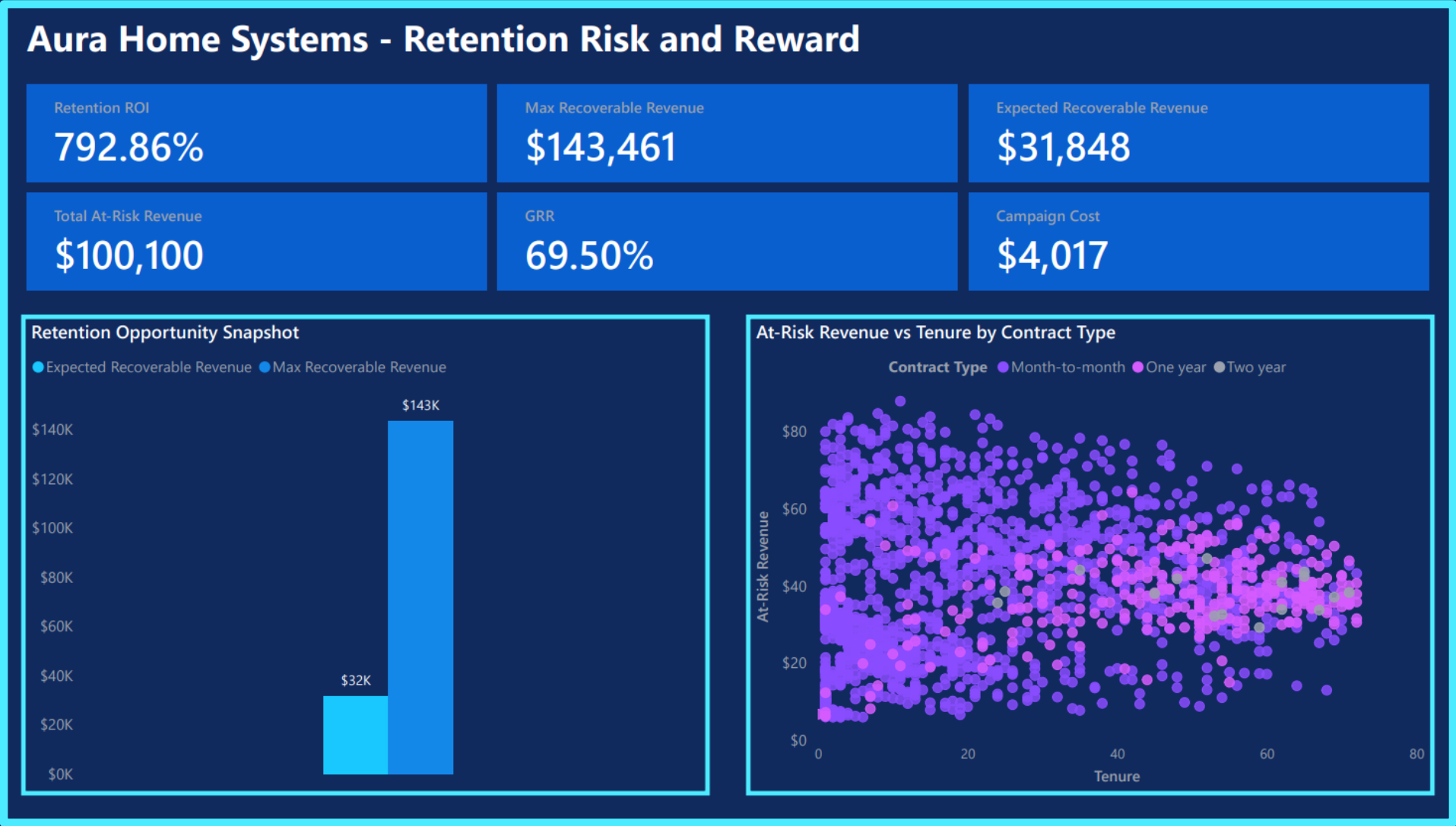


Churn Probability



“RETNETION RISK AND REWARD” DASHBOARD

An overall look at the risk vs reward of customer retention



"TIERS BY THE NUMBERS" DASHBOARD

Break down of tier data as a whole and by individual tier

Aura Home Systems - "Tiers" by the Numbers

Total "Tiers" Pot MRR Loss

\$143,461

Total "Tiers" At-Risk Revenue

\$77,257

Total "Tiers" Avg Churn Prob

53.50%

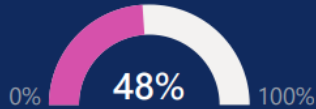
Total "Tiers" Avg CLV

\$351.62

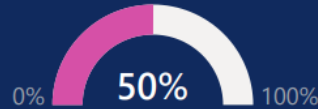
Total "Tiers" ARPU

\$61.27

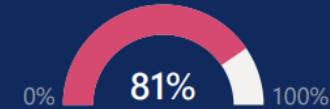
Tier 1 Churn Probability



Tier 2 Churn Probability

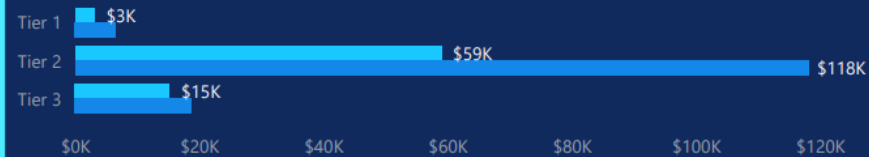


Tier 3 Churn Probability



At-Risk Rev vs Max Pot MRR Loss by Tier

● At-Risk Revenue ● Max Pot MRR Loss



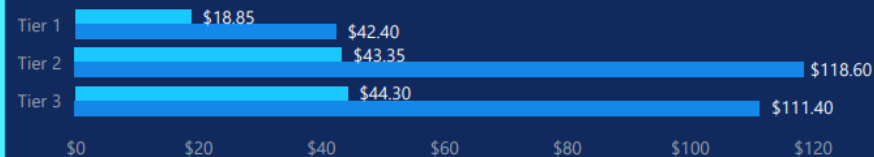
ARPU vs CLV by Tier

● ARPU ● CLV

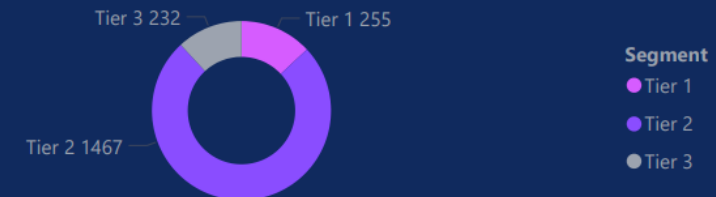


Min vs Max Monthly Charge by Tier

● Min Monthly Charge ● Max Monthly Charge

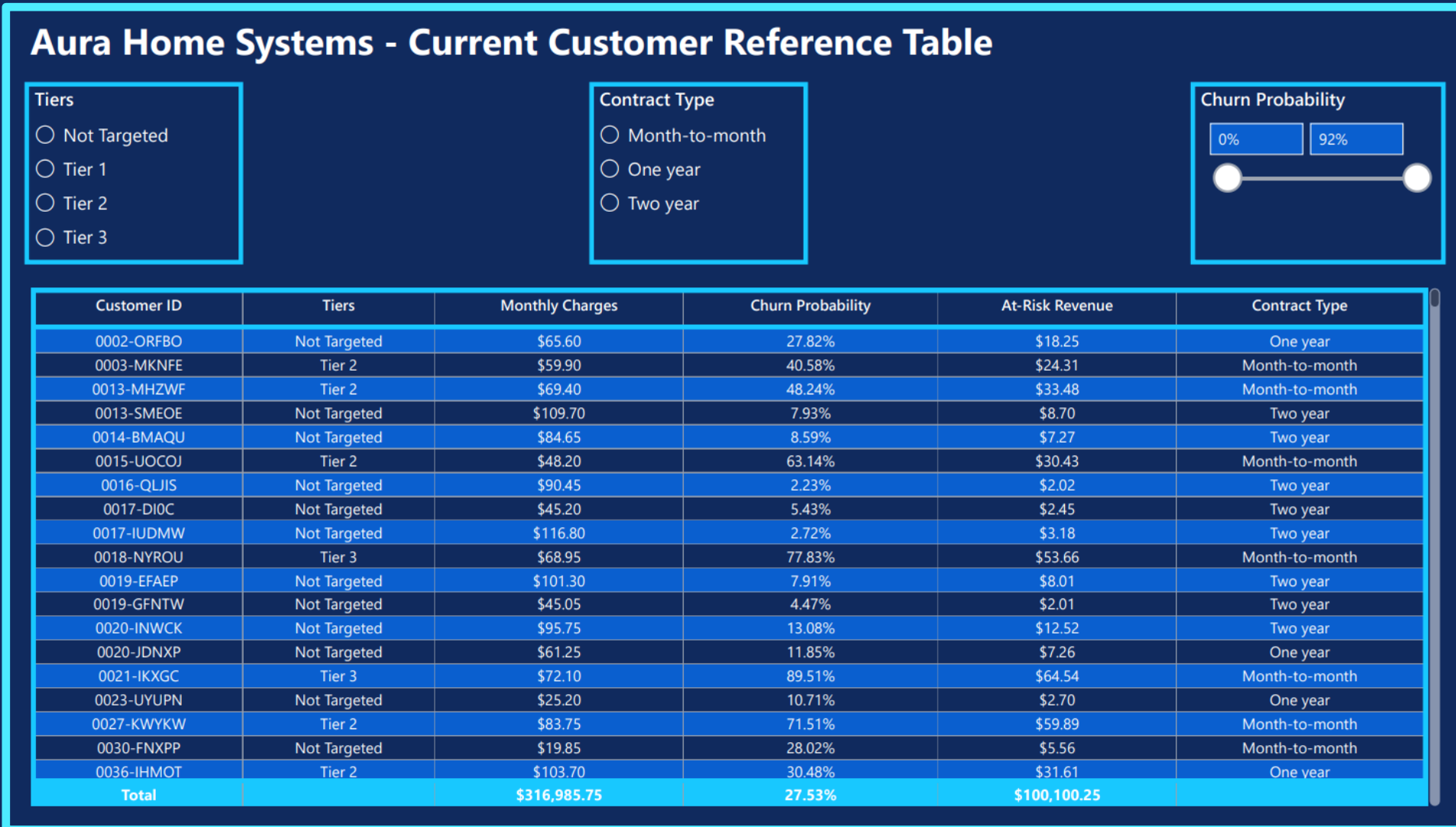


Customer Count by Tier



“CUSTOMER REFERENCE TABLE” DASHBOARD

A reference table for current customers with granular filtering capabilities.



CUSTOMER RETENTION APP

This app allows us to track customer behavior regarding retention efforts, specifically, if we were able to reach and/or retain them.

Rows shown are after required dataset filters are applied.

Filters

Search customer_id

gender

☒ All ☐ 0 ☐ 1

senior_citizen

☒ All ☐ 0 ☐ 1

partner

☒ All ☐ 0 ☐ 1

dependents

☒ All ☐ 0 ☐ 1

tenure

☒ 0.00 ☐ 72.00

0.00 72.00

phone_service

☒ All ☐ 0 ☐ 1

multiple_lines

☒ All ☐ 0 ☐ 1

internet_service

Aura Home Systems — Customer Retention

Upload your CSV, filter customers, and track outcomes with Available/Retained.

Upload retention dataset (CSV)

Drag and drop file here

Limit 200MB per file • CSV

Browse files

Aura_Tiers.csv 0.8MB

Legend: Any value of 0 means No. Any value of 1 means Yes.

Outcome Summary

Available

```
{
  "Yes" : 0
  "No" : 0
  "Blank" : 1954
  "Yes % (of answered)" : 0
  "No % (of answered)" : 0
}
```

Retained

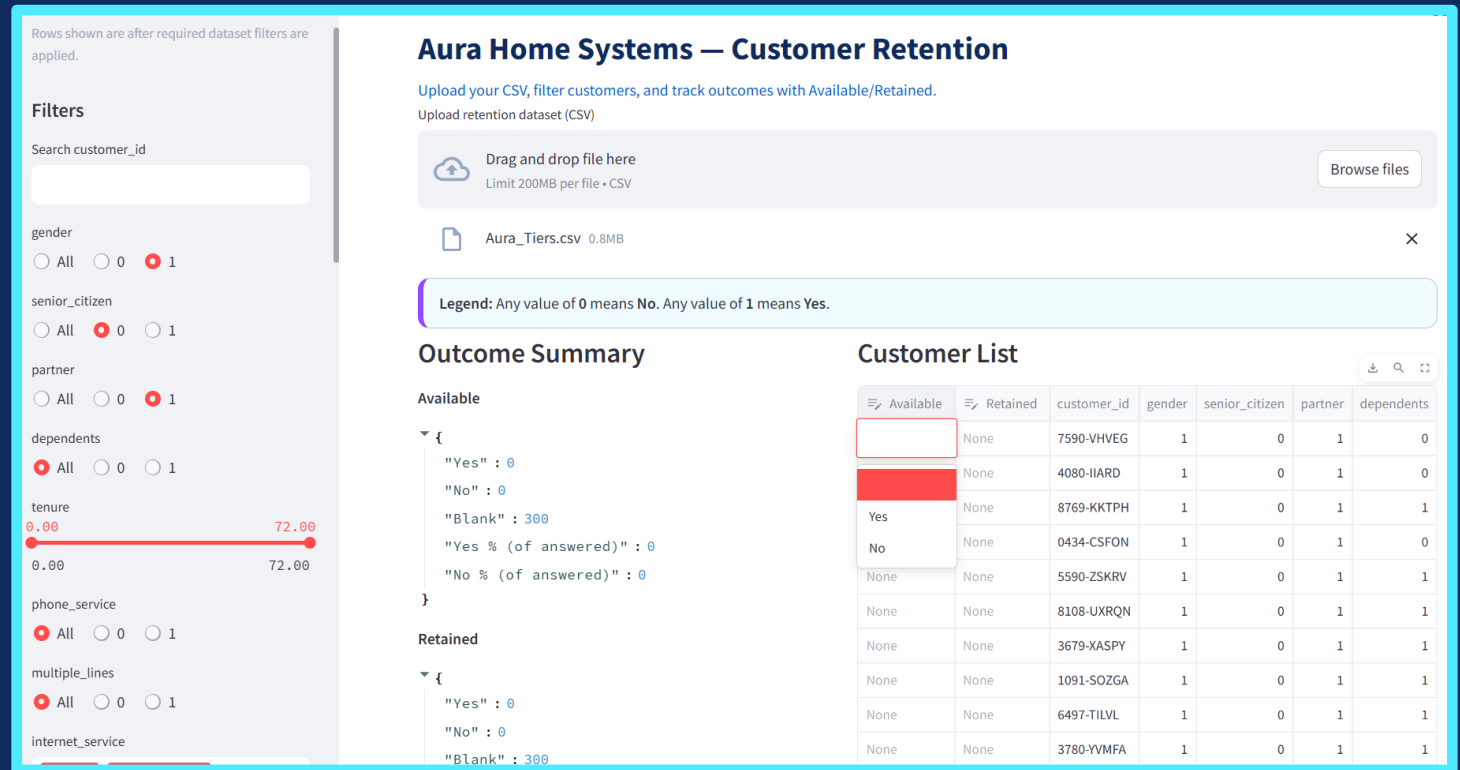
```
{
  "Yes" : 0
  "No" : 0
  "Blank" : 1954
}
```

Customer List

Available	Retained	customer_id	gender	senior_citizen	partner	dependents
None	None	7590-VHVEG	1	0	1	0
None	None	1452-KIOVK	0	0	0	1
None	None	6713-OKOMC	1	0	0	0
None	None	8091-TTVAX	0	0	1	0
None	None	5129-JLPIS	0	0	0	0
None	None	4183-MYFRB	1	0	0	0
None	None	4929-XIHVW	0	1	1	0
None	None	7310-EGVHZ	0	0	0	0
None	None	3413-BMNZE	0	1	0	0
None	None	6572-ADKRS	1	0	0	0

APP FUNCTIONALITY

- Full filtering capabilities using any fields within the dataset
- Tracking customer availability regarding retention efforts
- Tracking customer retention
- Summaries of availability and retention efforts



APP SUMMARY

We can use this app as a guide to see if our retention efforts are working, and if not, we can pivot and brainstorm new incentives for retention.

